Blessed G/Michael Catholic Secondary School

Semester 1 Physics Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** Which of the following is a pure vector quantity?
A. Speed **B.** Mass **C.** Displacement **D.** Kinetic energy

____ **2.** Two vectors of magnitudes 3 N and 4 N act at a right angle (90 degrees) to each other. What is their resultant magnitude?
A. 1 N **B.** 5 N **C.** 7 N **D.** 12 N

____ **3.** A car accelerates uniformly from rest to 20 m/s in 5 seconds. What is its acceleration?
A. 2 m/s^2 **B.** 4 m/s^2 **C.** 5 m/s^2 **D.** 10 m/s^2

____ **4.** How far does a body travel during 4 seconds of free fall from rest? (Take g = 10 m/s^2)
A. 40 m **B.** 60 m **C.** 80 m **D.** 100 m

____ **5.** In projectile motion, what happens to the horizontal velocity component (neglecting air resistance)?
A. It increases continuously
B. It decreases continuously
C. It remains constant throughout the flight
D. It becomes zero at maximum height

____ **6.** What launch angle produces the maximum horizontal range for a projectile launched on level ground?
A. 30 degrees **B.** 45 degrees **C.** 60 degrees **D.** 90 degrees

____ **7.** Newton’s First Law of Motion is also known as the Law of:
A. Acceleration **B.** Conservation of Momentum
C. Inertia **D.** Gravitation

____ **8.** A net force of 50 N acts on an object of mass 10 kg. What is the resulting acceleration?
A. 0.2 m/s^2 **B.** 5 m/s^2 **C.** 50 m/s^2 **D.** 500 m/s^2

____ **9.** Action and reaction force pairs mentioned in Newton’s Third Law:
A. Act on the same object and cancel each other
B. Act on two different interacting bodies
C. Are unequal in magnitude when there is acceleration
D. Act in the same direction

____ **10.** The coefficient of static friction (mu_s) compared to kinetic friction (mu_k) for the same surfaces is generally:
A. Greater **B.** Smaller **C.** Equal **D.** Zero

____ **11.** A 1500 kg truck travelling at 20 m/s brakes to a complete stop in 4 seconds. What magnitude of braking force was applied?
A. 3,000 N **B.** 5,000 N **C.** 7,500 N **D.** 15,000 N

____ **12.** What is the work done when a force of 40 N moves a box 5 meters in the exact direction of the force?
A. 8 J **B.** 45 J **C.** 200 J **D.** 800 J

____ **13.** A 1000 kg vehicle moves in a circle of radius 50 m at a speed of 10 m/s. What centripetal force is required?
A. 2,000 N **B.** 500 N **C.** 10,000 N **D.** 1,000 N

____ **14.** Which law of physics states that the total momentum of an isolated system remains constant in the absence of external forces?
A. Law of Conservation of Linear Momentum **B.** Newton's Law of Universal Gravitation
C. Hooke's Law **D.** Coulomb's Law

____ **15.** A ball is thrown vertically upward with initial velocity 30 m/s. How high does it rise? (g = 10 m/s^2)

A. 45 m **B.** 90 m **C.** 30 m **D.** 15 m

____ **16.** The slope of a velocity-time (v-t) graph represents:

A. Acceleration **B.** Displacement **C.** Speed **D.** Force

____ **17.** The area under a force-displacement (F-d) graph represents:

A. Work done (energy transferred) **B.** Power

C. Momentum **D.** Acceleration

____ **18.** What is the kinetic energy of a 2 kg object moving at a velocity of 6 m/s?

A. 36 J **B.** 12 J **C.** 72 J **D.** 18 J

____ **19.** If the net force acting on a moving object is suddenly reduced to zero, the object will:

A. Continue moving with constant velocity in a straight line
B. Immediately stop moving
C. Gradually slow down and stop
D. Accelerate rapidly

____ **20.** What is the weight of an 8 kg mass on the surface of the Moon? (g_moon = 1.6 m/s^2)

A. 12.8 N **B.** 80 N **C.** 50 N **D.** 0 N

____ **21.** A crate of mass 20 kg rests on a horizontal floor where coefficient of static friction mu_s = 0.4. What minimum horizontal force is needed to start it moving? (g = 10 m/s^2)

A. 80 N **B.** 200 N **C.** 50 N **D.** 8 N

____ **22.** Which of the following is true regarding work done by gravity on a satellite in a circular orbit around Earth?

A. Work done is zero because gravity is perpendicular to velocity
B. Work is positive
C. Work is negative
D. Work equals kinetic energy

____ **23.** Two vectors have equal magnitudes of 10 units each. If their resultant magnitude is also 10 units, what is the angle between them?

A. 120 degrees **B.** 60 degrees **C.** 90 degrees **D.** 180 degrees

____ **24.** What is the power delivered by an engine lifting a 400 kg elevator 10 m upward in 5 seconds at constant speed? (g = 10 m/s^2)

A. 8,000 W (8 kW) **B.** 4,000 W **C.** 40,000 W **D.** 2,000 W

____ **25.** An object has a momentum of 24 kg*m/s and a kinetic energy of 48 J. What is its mass?

A. 6 kg **B.** 4 kg **C.** 8 kg **D.** 12 kg